VEPOLINK
MONITORING PURITY

ENVIRONMENTAL INTELLIGENCE × GAS ANALYSIS

CONTINUOUS AMBIENT VOC MONITORING

Ambient BTEX Analyzer *GC-PID with Thermal Desorption*

Benzene, toluene, ethylbenzene and xylenes speciated every 30 minutes by gas chromatography with photo-ionization detection — pre-concentrated on Tenax and resolved to a tenth of a microgram per cubic metre.



0–20 ppbV
STANDARD MEASURING RANGE

30 min
ANALYSIS CYCLE

40 cmpds
CONFIGURABLE HYDROCARBON LIBRARY

GC-PID REFERENCE METHOD | **TENAX PRE-CONCENTRATION** TRACE ENRICHMENT | **CPCB** ALIGNED REPORTING

0.03 ppbV	<3 %	30 min	10 yr
LOWER DETECTABLE LIMIT, BENZENE	REPRODUCIBILITY AT 1 PPBV	FULL ANALYSIS CYCLE	EXPECTED SERVICE LIFE

01 OVERVIEW

Benzene and its neighbours, *speciated by name.*

A gas chromatograph with a built-in pre-concentration system. Ambient hydrocarbons enrich on a Tenax GR trap, desorb thermally, and separate on a capillary column before a photo-ionization detector — highly specific to benzene and other aromatics — reads each one out by name.

An industrial PC with a full-colour touchscreen runs the analysis and stores every chromatogram to disk, with RS-232, analog, digital and TCP/IP outputs so results reach a station, SCADA or a regulatory server automatically.

STANDARD REPORTED COMPOUNDS

- BENZENE
- TOLUENE
- ETHYLBENZENE
- O-XYLENE
- M-XYLENE
- P-XYLENE
- UP TO 40 CONFIGURABLE
- FULL CHROMATOGRAM
- VALIDATION STATUS
- FAULT DIAGNOSTICS

· PRODUCT FEATURES

Photo-Ionization Detector

A PID gives highly specific sensitivity to benzene and aromatic hydrocarbons generally, distinguishing them cleanly from other species.

Tenax Pre-Concentration

Trace-level ambient hydrocarbons enrich on a Tenax GR trap before desorption, which is what makes sub-µg/m³ detection practical.

Automatic Validation

Software-driven multipoint validation checks the calibration using varying trap volumes, without manual intervention.

Ten Years of Design Life

Low maintenance costs, a yearly preventive service and economically priced consumables keep the analyzer running for a decade.

02 MEASUREMENT PRINCIPLE

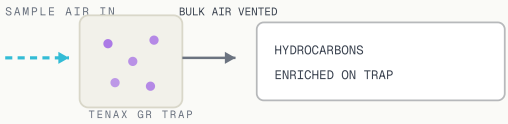
How gas chromatography *separates BTEX.*

Ambient hydrocarbons arrive as a mixture at parts-per-billion levels. Enrichment, separation and a compound-specific detector turn that mixture into a named, quantified list every cycle.

PRE-CONCENTRATION

Tenax GR Trap

SORBENT TRAP · CONTROLLED SAMPLE VOLUME



Sample air is drawn through a Tenax GR sorbent trap. Hydrocarbons adsorb and enrich on the trap while the bulk of the air passes through and vents.

`Trace levels, enriched`

SEPARATION

Thermal Desorption & Column

FLASH HEATING · CAPILLARY COLUMN



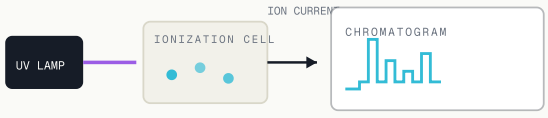
The trap is flash-heated; desorbed hydrocarbons sweep onto a capillary column and separate by retention time into resolved, named peaks.

`One mixture, many peaks`

DETECTION

Photo-Ionization Detector

UV IONIZATION · ION CURRENT



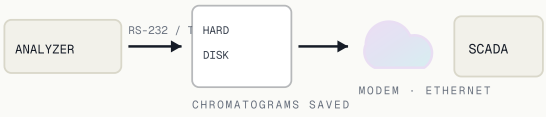
Eluting compounds are ionized by UV light; the resulting current is proportional to concentration, with high specificity for aromatics.

`Ion current ∝ concentration`

DATA PATH

Chromatogram & Reporting

RS-232 · TCP/IP · ANALOG / DIGITAL



Every run's chromatogram is saved to disk and identified by retention time, then reported over network, modem or serial connection.

`Reviewable, exportable data`

03 ANALYSIS CYCLE

One 30-minute run, *start to result.*

Each cycle repeats the same four steps automatically, producing a fresh chromatogram without operator intervention.

STEP 01

Trap loading

Ambient air draws through the Tenax GR trap at a controlled volume; hydrocarbons adsorb as bulk air vents.

STEP 02

Thermal desorption

The trap flash-heats; desorbed hydrocarbons sweep by nitrogen carrier gas onto the capillary column.

STEP 03

Column separation

Compounds separate by retention time; benzene, toluene, ethylbenzene and xylene isomers elute individually.

STEP 04

Detect and report

The PID measures each peak, retention time identifies the compound, and the chromatogram is stored and transmitted.

WHY PRE-CONCENTRATION MATTERS

Ambient benzene runs from under 0.1 to over 100 µg/m³ — far below what a detector could read directly at reasonable sample volumes. Trapping and enriching the sample before desorption is what makes a lower detectable limit of 0.03 ppbV achievable in the field, not just on the bench.

EXTENDING THE COMPOUND LIST

The same hardware, with different settings and calibration gas, resolves up to 40 hydrocarbons in one run — octane, chloroethenes and trimethylbenzenes among them. Tell us which compounds matter for your site and we configure the peak library accordingly.

WHY BENZENE IS MONITORED

Benzene is a known carcinogen with no absolute safe concentration — CPCB accordingly makes its ambient measurement obligatory, with an annual mean limit under 5 µg/m³. It reaches ambient air mainly through evaporated fuel at filling stations and from vehicle exhaust, with toluene, ethylbenzene and xylene isomers arriving from the same combustion and evaporative sources.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

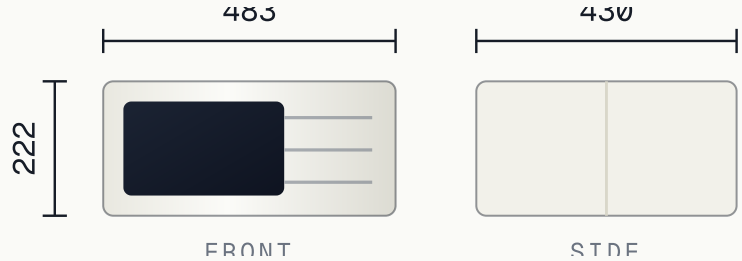
Parameter	Benzene, toluene, ethylbenzene, xylenes + up to 40 hydrocarbons
Working Principle	GC with PID, Tenax pre-concentration & thermal desorption
Standard Range	0 - 20 ppbV (up to 300 ppbV with software)
Lower Detectable Limit	0.1 µg/m³ (0.03 ppbV), benzene
Reproducibility	< 3% at 1 ppbV, benzene, capillary column
Standard Calibration	4-point, 4 - 16 ppbV
Analysis Cycle	30 minutes per run
Compound Library	Up to 40 hydrocarbons, configurable
Carrier Gas	Nitrogen, quality 5.0, 4 bar, 6 ml/min
Expected Lifetime	10 years

SYSTEM · COMMS · BUILD

Display	Full-colour touchscreen
Controller	Industrial x86 PC with hard disk storage
Software	Windows embedded + GC955
Control	Touchscreen, keyboard, mouse or remote host
Interfaces	RS-232 · analog & digital I/O via TCP/IP
Protocols	ASCII terminal · Hessen · Gesytec · MODBUS
Data Storage	Chromatograms saved to disk, up to 10 years
Power Supply	230 VAC, 100 VA, 50/60 Hz (115 VAC available)
Mounting	19-inch rack, 5 standard height units
Weight	19 kg

INSTRUMENT DIMENSIONS

INDICATIVE FRONT & SIDE ELEVATION · MM



05 INSTALLATION & UPKEEP

Racked, calibrated, *left running.*

Ambient benzene monitoring is obligatory under CPCB regulation, with an **annual mean limit under 5 µg/m³**. The analyzer is installed in a 19-inch, 5U rack position with a nitrogen carrier gas supply and a scheduled **multipoint validation** against certified calibration gas.

01 · MOUNTING

19-inch rack, 5U

Standard rack fitment alongside other CAAQMS analyzers, sharing the station's power and network connection.

02 · SAMPLE LINE

Inert, direct to trap

A short inert line delivers ambient air to the Tenax GR trap inlet at the controlled sampling volume.

03 · CARRIER GAS

Nitrogen, 4 bar

A quality-5.0 nitrogen cylinder, regulated to 4 bar, supplies carrier gas at roughly 6 ml/min.

04 · CALIBRATION

4-point + auto validation

A standard 4-point calibration (4–16 ppbV) is backed by automatic multipoint validation using varying trap volumes.

05 · CONSUMABLES

Trap, column, carrier gas

Economically priced Tenax traps and capillary columns, plus the carrier gas cylinder, are the routine consumables.

06 · DIAGNOSTICS

Reported with the data

Validation status, trap condition and chromatogram quality travel alongside every run.

PART OF A FULL STATION

The BTEX analyzer is normally specified alongside SO_2, NOx, CO and ozone analyzers to give a station full gaseous-pollutant coverage, including the aromatic VOCs those instruments don't speciate. Sharing power and a data path lets it report into the same station record.

COMMISSIONING & AUDIT

Commissioning runs the standard 4-point calibration, confirms peak identification by retention time for every target compound, and establishes a validation baseline. Preventive maintenance is required only once a year; the automatic validation routine carries day-to-day confidence between services.

06 APPLICATIONS

Where it *works.*

Wherever fuel is stored, dispensed or burned, aromatic VOCs need speciated tracking — not just a total hydrocarbon number.

Regulatory CAAQMS Networks	Fuel Filling Stations	Refineries & Petrochemicals
Urban & Traffic Corridors	Chemical Plants	Ports & Fuel Terminals
Storage Tank Farms	EIA & Baseline Studies	Research & VOC Studies

WHAT THE DATA SUPPORTS

Speciated BTEX data serves regulators tracking benzene against the annual CPCB limit, industry teams isolating a fuel-handling or process source from background traffic, and communities near a filling station or terminal who want a specific, named answer rather than a lumped VOC total.

SITING NEAR A SOURCE

Fence-line installations near filling stations, terminals or tank farms are placed downwind of the prevailing direction, with a background station for comparison. Because each cycle takes 30 minutes, siting favours locations where a half-hourly average is the meaningful unit — regulatory and source-attribution studies, not instantaneous leak detection.

07 FAQ

Frequently asked *questions.*

What is the BTEX analyzer?

An online gas chromatograph that continuously measures benzene, toluene, ethylbenzene and xylenes (BTEX) in ambient or industrial air, one full cycle every 30 minutes.

Why is BTEX monitoring important?

These are volatile organic compounds with real health and environmental impact — benzene in particular is carcinogenic — so monitoring supports air quality compliance and public safety.

How does it detect BTEX compounds?

Pre-concentration on a Tenax GR trap, thermal desorption, capillary column separation and photo-ionization detection give precise, continuous quantification of each compound by name.

Where is it used?

Environmental monitoring stations, industrial emission monitoring, fuel filling stations, petrochemical facilities and research labs.

Does it support remote data output?

Yes — remote connectivity and data export to SCADA systems, servers or cloud platforms for real-time tracking, via RS-232, analog, digital and TCP/IP.

Can it measure compounds beyond BTEX?

Yes. With different settings and suitable calibration gas it can resolve up to 40 hydrocarbons in one run — tell us the compound list and we check the configuration.

How sensitive is it to benzene?

The lower detectable limit is 0.1 µg/m³ (0.03 ppbV), with reproducibility under 3% at 1 ppbV using a capillary column.

How is it calibrated?

A standard 4-point calibration covers 4–16 ppbV, and the software runs automatic multipoint validation using different sample volumes through the trap.

What does it need to run?

A small quantity of nitrogen carrier gas (about 6 ml/min from a quality-5.0 cylinder at 4 bar) and standard 230 VAC power.

What maintenance does it need?

Preventive maintenance only once a year. Consumables — the Tenax trap and capillary column — are economically priced, and the expected instrument lifetime is 10 years.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. For installation guidelines and full technical detail, refer to the user manual supplied with the product. Specifications may change without notice as the product is improved.

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AROMATIC HYDROCARBONS, NAMED AND QUANTIFIED

Know your BTEX. *Compound by compound.*

Tell us the site, the compound list you need and who receives the data, and we'll configure the analyzer around it — calibration gas, carrier gas supply, rack space and reporting. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that specifies, installs and calibrates ambient gas analyzers. No call centre.
+91 99538 86178 →

02 **Request a station configuration**
RECOMMENDED
Send your site and compound requirement — we'll spec calibration, carrier gas, rack space and communications.
sales@vepolink.com →

03 **Book a demonstration**
See a running analyzer and a live chromatogram with our engineering team.
Request a demo →

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